

```

    name: "{{ item }}"
    state: latest
with_items:
  - openssl
  - python-openssl

- name: Generate an OpenSSL private key
  openssl_privatekey:
    path: /etc/ssl/private/ansible.com.pem

- name: Create directory
  file:
    path: /etc/ssl/csr
    state: directory
    mode: 0755

- name: Generate an OpenSSL Certificate Signing Request
  openssl_csr:
    path: /etc/ssl/csr/www.ansible.com.csr
    privatekey_path: /etc/ssl/private/ansible.com.pem
    country_name: IN
    organization_name: Ansible
    email_address: author@shakthimaan.com
    common_name: www.ansible.com

- name: Generate a self signed certificate
  openssl_certificate:
    path: /etc/ssl/certs/nginx-selfsigned.crt
    privatekey_path: /etc/ssl/private/ansible.com.pem
    csr_path: /etc/ssl/csr/www.ansible.com.csr
    provider: selfsigned

```

The above playbook can be run as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/nginx-ssl.yml --tags ssl -K
```

Configuring Nginx for SSL

The final step is to configure Nginx to use SSL. A *self-signed.conf* file is created in the */etc/nginx/snippets* folder that contains the following:

```
ssl_certificate /etc/ssl/certs/nginx-selfsigned.crt;
ssl_certificate_key /etc/ssl/private/ansible.com.pem;
```

The SSL parameter configurations are stored in the */etc/nginx/snippets/ssl-params.conf* file as shown below:

```
# from https://ciphersli.st/
# and https://raymii.org/s/tutorials/Strong_SSL_Security_On_
nginx.html

ssl_protocols TLSv1 TLSv1.1 TLSv1.2;
ssl_prefer_server_ciphers on;
```

```
ssl_ciphers "EECDH+AESGCM:EDH+AESGCM:AES256+EECDH:AES256+E
DH";
ssl_ecdh_curve secp384r1;
ssl_session_cache shared:SSL:10m;
ssl_session_tickets off;
ssl_stapling on;
ssl_stapling_verify on;
resolver 8.8.8.8 8.8.4.4 valid=300s;
resolver_timeout 5s;
# Disable preloading HSTS for now. You can use the commented
out header line that includes
# the "preload" directive if you understand the implications.
#add_header Strict-Transport-Security "max-age=63072000;
includeSubdomains; preload";
add_header Strict-Transport-Security "max-age=63072000;
includeSubdomains";
add_header X-Frame-Options DENY;
add_header X-Content-Type-Options nosniff;
```

The Nginx Web server configuration for this host (*google.com*, for example) is then created in the */etc/nginx/sites-enabled* folder with the following contents:

```
server {
    listen 80;

    root /var/www/html;
    index index.nginx-debian.html;

    server_name google.com www.google.com;
}

server {
    listen 443 ssl http2 default_server;
    include snippets/self-signed.conf;
    include snippets/ssl-params.conf;
}
```

The Ansible playbook for configuring Nginx with SSL is as follows:

```
- name: Setup nginx with SSL
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [https]

tasks:
  - copy:
      src: ../../files/self-signed.conf
      dest: /etc/nginx/snippets/self-signed.conf
      owner: root
      group: root
      mode: 0644
```